SYRIAN PRIVATE UNIVERSITY

Faculty of Dentistry

SPU Dental Clinic System

Software Requirements Specification

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This specification defines what the system must do. It is written for approval by the dean of the Faculty of Dentistry and as the build input for the development team.

1. Introduction

Purpose. This document states the requirements for the SPU Dental Clinic System, which replaces the paper forms of the Faculty of Dentistry clinics and labs at Syrian Private University. It has three readers. The dean approves it. The development team builds from it. Claude Code uses its requirement IDs and acceptance criteria as build instructions.

Scope. The system has:

- one mobile app (iOS and Android) with three interfaces: Student, Supervisor and Maintenance
- one web dashboard for the super admin and management
- one Django backend that serves both

It covers patient records, diagnosis, referrals, clinical and lab work cards with grading, chair and session distribution, chair checks and maintenance, photos and reports. It does not cover fees or payments, the university's academic system, or the X-ray imaging software.

Definitions

Term	Meaning
Course (مقرر)	A clinical or lab course a student is enrolled in for a term. It defines the required cases and the steps of each case
Work card (بطاقة الأعمال)	The electronic version of F2 (lab) or F3 (clinical): one per student per course per term
Case	One unit of work on a work card, for example one filling on tooth 36 or one lab exercise
Step	One stage of a case (for example Preparation), approved and graded by a supervisor
Diagnosis	The electronic version of F1: history, dental chart, examination, diagnosis and treatment plan
Plan item	One treatment in the diagnosis plan, for example a root canal on tooth 46
Referral (إحالة)	Sends one or more plan items to a department clinic for treatment
Clinic	A room with chairs (clinical) or benches (lab). Its type is diagnosis, treatment or lab
Session	A time slot in the clinic day, for example 8:00–9:40
Round	A morning or afternoon chair check (F6, F18)
Ticket	A fault report on a chair, followed until it is repaired
Super admin	The dashboard user who manages the whole system and registers patients
FDI	The two-digit tooth numbering used on F1 (quadrant 1–4, tooth 1–8)

References

- SPU Dental Clinic System — System Analysis (team document)
- SPU Dental Clinic System — Project Plan (team document)

- SPU/AQM forms M-09-Den-F1, F2, F3, F6, F7, F8, F9, F18 (issue 00, 01/11/2025)
- Site visit recording, clinics, 18 Oct 2026

2. Overall description

The system is new and stands alone. One Django backend holds all data. It serves the mobile app through a REST API and the dashboard directly.

Product perspective

- **Backend:** Django, Django REST Framework, PostgreSQL. It holds all records, the rules and the permissions.
- **Dashboard:** Django Unfold, inside the same Django project. Only the super admin and management use it.
- **Mobile app:** Flutter. One app; after login it opens the Student, Supervisor or Maintenance interface according to the user's role. A user with several roles picks one.
- **Storage:** photos and X-ray images in S3-compatible object storage.
- **Notifications:** in-app list for every user, plus push notifications where the device supports them.

User classes

User class	Interface	Typical number	Main need
Super admin	Dashboard	1–3	Set up and run the whole system; register patients; manage users; view everything
Management (dean, department heads)	Dashboard, read-only	5–15	Reports, statistics, referral and maintenance overview
Student	Mobile, Student interface	Hundreds (to confirm)	Diagnose patients, record treatment steps, follow own work cards and grades
Supervisor	Mobile, Supervisor interface	Tens (to confirm)	Approve and grade, refer patients, assign patients to students, follow attendance
Maintenance technician	Mobile, Maintenance interface	1–5	Chair rounds, fault tickets, repairs

Operating environment

- **Mobile:** Android 8.0 or later; iOS 15 or later; phones and tablets.
- **Dashboard:** current versions of Chrome, Edge, Firefox and Safari on desktop.
- **Server:** Linux cloud server with Docker, HTTPS, on the university domain.
- **Time zone:** Asia/Damascus.

Constraints

- Arabic first, written right to left. English is available as a second language.
- Nothing about clinics, chairs, courses, steps or sessions may be fixed in code. All of it is configured in the dashboard (section 4).
- There are no fees or payments.

- The content of every SPU/AQM paper form must be covered by a screen.
- Fixed handover date: 18 Jan 2027.

Assumptions and dependencies

- Student, course and enrollment lists come from the academic system as Excel files.
- The app is published from developer accounts held outside Syria.
- The university provides a subdomain for the server.
- An electronic approval by a logged-in supervisor replaces the handwritten signature on the forms.

3. Roles and permissions

The server checks every action against the user's role and, for students, against ownership. A student only ever sees patients and cases assigned to them.

Key: ✓ allowed · *own* only their own records · *scope* only clinics or courses the supervisor is assigned to · *view* read-only · — not allowed

Action	Super admin	Management	Supervisor	Student	Technician
Configure clinics, chairs, sessions, courses, steps, lists	✓	—	—	—	—
Manage users and import students	✓	—	—	—	—
Register and edit patients	✓	—	—	—	—
View a patient record	✓	view	scope	own	—
Assign a waiting patient to a student for diagnosis	✓	—	scope	—	—
Fill and submit a diagnosis	—	—	—	own	—
Approve or return a diagnosis	—	—	scope	—	—
Create a referral	—	—	scope	—	—
Accept or reject a referral, assign to a student	✓	—	scope	—	—
Record a step and upload photos	—	—	—	own	—
Create a lab case	—	—	—	own	—
Approve, grade or return a step	—	—	scope	—	—
Transfer a patient between students	✓	—	scope	—	—
Distribute students to chairs and sessions	✓	—	scope	view own	—
Mark attendance	—	—	scope	—	—
Do a chair round	—	—	—	—	✓
Report a chair fault	✓	—	scope	—	✓
Work and close a ticket; set a chair in or out of service	✓	—	—	—	✓

Action	Super admin	Management	Supervisor	Student	Technician
View reports and statistics	✓	view	scope	own	own tickets
View the audit log	✓	—	—	—	—

A user can hold several roles, for example a supervisor who is also a department head (Management). Permissions add up across roles.

4. Dynamic configuration

Everything the clinic may change is data that the super admin edits in the dashboard, never code. New clinics, chairs, courses or steps take effect without a new release of the app.

Item	Fields the admin sets	Used by
Academic term	Name (for example 2026–2027 Fall), start and end dates, active flag	Courses, enrollments, work cards
Department	Name (AR and EN), head (user)	Clinics, courses, referrals, reports
Clinic	Number, name, building and floor, type (diagnosis / treatment / lab), department, active flag	Distribution, referrals, rounds, reports
Chair or bench	Clinic, number, status (in service / out of service)	Distribution, rounds, tickets
Session	Clinic type or clinic, name, start and end time, order	Distribution (F7), daily reports (F8, F9)
Course	Code, name (AR and EN), type (clinical / lab), department, term, clinics where it runs, students per chair (1–3)	Work cards, assignment, distribution
Case requirement	Course, case type, number required	Work card progress, case finder
Step template	Course, case type (optional), order, name, grading type, photo required (yes / no)	Cases, approvals
Grading type	Numeric score with a range (for example 0–100), a code list (for example A/B/C), or pass / fail	Step approval
Course supervisors	Course or clinic, supervisors	Supervisor scope (section 3)
Enrollment	Student, course, term (usually imported from Excel)	Work cards, eligibility
Tooth codes	Letter, meaning (AR and EN), colour on the chart. Default: D, M, F, N, R, E, C, I, U, P	Dental chart
Medical history items	Question (AR and EN), type (yes/no, yes/no + details, text), alert flag, order. Default: the 15 conditions on F1, plus smoking, pregnancy, medications and past surgeries	Diagnosis, patient alerts
Soft tissue sites	Name (AR and EN), order. Default: the sites on the F1 oral map	Diagnosis

Item	Fields the admin sets	Used by
Periodontal index set	Index name (calculus / plaque), teeth and surfaces (default 16, 11, 26 buccal; 36, 31, 46 lingual), score range (default 0–3)	Diagnosis
Treatment types	Name (AR and EN), department it is referred to, case type it becomes	Treatment plan, referrals
Chair checklist	Item name (AR and EN), order. Default: cleanliness, handpiece connections, chair movements, lighting	Rounds
Round times	Morning and afternoon windows. Default: 8:30–9:30 and 14:00–15:00	Rounds, reminders
Fault categories	Name, blocks the chair (yes / no)	Tickets
System settings	University and faculty names, logo, default language, photo size limit, notification switches	Everywhere

The defaults listed come from the SPU/AQM forms. They are loaded on first install, and the admin can change them.

5. Functional requirements

There are 67 requirements across the dashboard (DB), the Student (ST), Supervisor (SV) and Maintenance (MT) interfaces, and the system as a whole (SYS). Priority uses MoSCoW: **Must** for the handover, **Should** if time allows, **Could** later. The sprint column matches the Project Plan.

5.1 Web dashboard (super admin and management)

ID	Requirement	Priority	Sprint
DB-01	Log in with username and password; management users get read-only menus	Must	S1
DB-02	Create, edit and deactivate users; assign one or more roles and a department; reset passwords	Must	S1
DB-03	Import students and enrollments from an Excel template, with a validation report of rejected rows	Must	S1
DB-04	Manage terms, departments, clinics, chairs, sessions and courses (section 4)	Must	S1
DB-05	Manage case requirements, step templates and grading types per course	Must	S1
DB-06	Manage lookup lists: tooth codes, medical history items, soft tissue sites, index set, treatment types, checklist items, fault categories	Must	S1
DB-07	Assign supervisors to courses and clinics	Must	S1
DB-08	Register a patient: full name, national ID, date of birth, sex, job, phone, address. The system generates a unique file number and warns about duplicates by national ID or by name plus phone	Must	S3
DB-09	Send a registered patient to a diagnosis clinic's waiting queue	Must	S3

ID	Requirement	Priority	Sprint
DB-10	Search patients by file number, national ID, name or phone; open the full record (read-only for clinical parts)	Must	S3
DB-11	Edit patient details, with a change history	Must	S3
DB-12	Distribute students to chairs and sessions for a clinic and a date range (F7), with an auto-suggest that respects students per chair and skips out-of-service chairs	Should	S4
DB-13	Referral board: all referrals with status, age in days and filters by clinic, department and status	Must	S4
DB-14	Maintenance board: chair status per clinic, open tickets, repair history	Must	S4
DB-15	Reports: daily clinic report (F8), daily lab report (F9), chair status (F6/F18), student logbook, course progress, patients with pending plan items, referral and maintenance statistics; export to Excel and PDF	Must	S5
DB-16	Home page with key figures: patients registered, open referrals, approvals waiting, chairs out of service	Should	S5
DB-17	Audit log viewer with filters by user, record and date	Must	S5
DB-18	System settings (section 4)	Must	S1
DB-19	Merge duplicate patient records	Could	—

5.2 Student interface (mobile)

ID	Requirement	Priority	Sprint
ST-01	Log in; change password on first login; stay logged in until logout or 30 days of inactivity	Must	S1
ST-02	Home: today's clinic, chair and sessions; returned items to fix; items waiting for approval; latest notifications	Must	S4
ST-03	My patients: patients assigned to me for diagnosis or treatment, with medical alerts shown first	Must	S3
ST-04	Patient record (own patients only): details, alerts, diagnoses, plan and its status, referrals, photos and X-rays	Must	S3
ST-05	Fill a diagnosis (F1) as a step-by-step form: complaint; medical history; dental chart (tap a tooth, pick codes); soft tissue sites; calculus and plaque indices; X-ray findings with images; diagnosis; treatment plan items (treatment type, teeth). Saves a draft automatically	Must	S3
ST-06	Submit the diagnosis to a supervisor of that clinic; see a returned diagnosis with the comment, fix it and resubmit	Must	S3
ST-07	My work cards: one card per course and term, showing cases done against required, and grades	Must	S2
ST-08	Case view: patient, tooth, case type and the list of steps with status, grade and supervisor	Must	S2
ST-09	Record a step: mark it done, add a note, attach photos from the camera or gallery, and send it to a supervisor for approval	Must	S2

ID	Requirement	Priority	Sprint
ST-10	Create a lab case on a lab work card: case type and exercise or model number	Must	S2
ST-11	Upload photos compressed, with a queue that retries when the network returns	Must	S2
ST-12	My schedule: chair and session distribution for the coming days	Should	S4
ST-13	Logbook: all my cases by course and case type, with grades	Should	S5
ST-14	Notifications list; tap to open the related item	Must	S5
ST-15	Profile, Arabic / English switch, logout	Must	S1

5.3 Supervisor interface (mobile)

ID	Requirement	Priority	Sprint
SV-01	Log in, same as ST-01	Must	S1
SV-02	Home: approvals waiting, by type; my clinics today; new referrals; patients waiting for diagnosis	Must	S4
SV-03	Diagnosis queue: assign a waiting patient to a student in my clinic	Must	S3
SV-04	Review a diagnosis (all sections, photos, X-rays), then approve it or return it with a comment	Must	S3
SV-05	From an approved diagnosis, select plan items and create a referral to a department clinic, with the procedure and a note	Must	S3
SV-06	Referral inbox for my clinics: accept or reject with a reason; assign the patient to an eligible student (enrolled in a course run in that clinic this term), with the student's progress shown	Must	S4
SV-07	Approval queue for steps: see the photos and the student's note; enter a grade according to the course's grading type; approve or return with a comment. Two taps for a normal approval	Must	S2
SV-08	When the last step is approved, confirm the case as complete. The referral then closes and the referring clinic is notified	Must	S4
SV-09	Transfer a patient (and their open cases) to another student, with a reason	Must	S4
SV-10	Case finder: search patients with pending plan items by treatment type, department or tooth, to give students the cases they need	Should	S4
SV-11	Clinic today: chair × session grid of students; mark attendance; adjust the distribution	Should	S4
SV-12	View a student's work cards and progress (students in my scope)	Should	S5
SV-13	Report a chair fault with a photo and a note; it opens a ticket	Must	S4
SV-14	Notifications list	Must	S5

5.4 Maintenance interface (mobile)

ID	Requirement	Priority	Sprint
MT-01	Log in, same as ST-01	Must	S1
MT-02	Home: clinics with counts of chairs in and out of service; open tickets by age	Must	S4
MT-03	Do a round: pick a clinic and the morning or afternoon round, then check each chair against the checklist. "All OK" in one tap; a fault takes a note and an optional photo	Must	S4
MT-04	A failed check opens a ticket automatically, linked to the chair and the round	Must	S4
MT-05	Work a ticket: Open → In progress → (External repair) → Resolved; mark it internal or external; add repair notes and photos	Must	S4
MT-06	A ticket in a blocking fault category takes the chair out of service. Resolving it returns the chair to service	Must	S4
MT-07	Set a chair in or out of service by hand, with a reason	Must	S4
MT-08	Chair history: past rounds, tickets and repairs	Should	S5
MT-09	Reminder notification when a round has not been done by the end of its window	Could	S5

5.5 System-wide

ID	Requirement	Priority	Sprint
SYS-01	JWT login with refresh tokens; lock an account for 15 minutes after 5 failed attempts	Must	S1
SYS-02	Role and ownership checks on every API call (section 3), never only in the app	Must	S1
SYS-03	In-app notifications for: diagnosis submitted, approved or returned; step submitted, approved or returned; referral received, accepted, rejected or completed; patient assigned or transferred; ticket opened or resolved. Push notifications for the same events where available	Must	S5
SYS-04	Audit log of every create, change, approval and login: user, action, record, time	Must	S1
SYS-05	Approved diagnoses and steps are locked. A change after approval creates a new version, and the history is kept	Must	S3
SYS-06	Photos: JPEG or PNG, compressed on the phone to at most 2 MB, with thumbnails generated on the server; linked to the patient and optionally to a case and step	Must	S2
SYS-07	Arabic and English for every screen, label and lookup list, right to left in Arabic	Must	S1
SYS-08	Printable PDF of a diagnosis (F1 layout), a work card (F2/F3) and the daily reports	Should	S5
SYS-09	API documented with OpenAPI at /api/v1/docs	Must	S1

ID	Requirement	Priority	Sprint
SYS-10	Soft delete only: records are deactivated, never removed, so history and grades stay intact	Must	S1

6. Business rules and record lifecycles

Four records carry the workflow. Each one moves forward only through a named person's action, and the server rejects any change that skips a state.

[embedded content: record lifecycles · diagnosis, referral, step, ticket\]

A return sends the record back to the student to fix and resubmit. A diagnosis can be returned many times before it is approved.

Business rules

ID	Rule
BR-01	Only the super admin creates patients. Each patient has a unique file number generated by the system and a unique national ID when one is given
BR-02	A patient has at most one diagnosis in progress at a time. A later diagnosis is a new record; the earlier ones stay as history, and the medical history is copied forward to be checked
BR-03	Students see and work only on patients assigned to them, for diagnosis or for treatment. A transfer ends the old student's access to that patient's open cases
BR-04	An approved diagnosis is locked. Only its plan items can be referred, and each item can be in only one open referral at a time
BR-05	A referred patient may be assigned only to a student enrolled, in the current term, in a course that runs in the receiving clinic
BR-06	Steps of a case are approved in their template order unless the course allows any order (configurable)
BR-07	A supervisor approves only within their scope (assigned courses or clinics), and never their own work
BR-08	When the last step of a case is approved and the case is confirmed complete, the case is marked done, its referral is completed, the plan item is marked done, and the referring clinic is notified
BR-09	A case counts toward a work card's required number only when it is complete. Work done on a transferred patient stays credited to the student who did each step
BR-10	A lab work card needs the course's required number of lab cases (default 5, from F2)
BR-11	One round per clinic, round type (morning / afternoon) and day. After submission, only the super admin can change it
BR-12	A failed check opens a ticket. A ticket in a blocking category takes the chair out of service; out-of-service chairs cannot be given to students
BR-13	Nothing is deleted: records are deactivated, and approved records are versioned (SYS-05, SYS-10)
BR-14	Patients pay no fees, and the system records no payments

7. User stories and acceptance criteria

There are 24 user stories, grouped by the sprint that delivers them. Each story is done when every acceptance criterion passes, together with the definition of done in the Project Plan.

Sprint 1 — Foundation

ID	Story	Acceptance criteria	Covers
US-01	As the super admin, I want to set up a clinic with its chairs and sessions, so that the system matches the real faculty	Create clinic 5 (treatment, 12 chairs) and 5 sessions. The chairs appear in distribution and rounds. Deactivating a chair hides it from new distributions but keeps its history	DB-04
US-02	As the super admin, I want to define a course with its case requirements and steps, so that work cards follow the course rules	Create a course with 2 case types, a required count for each, 4 steps and a 0–100 grading type. A student enrolled in it gets a work card showing 0 of the required count	DB-04, DB-05
US-03	As the super admin, I want to import students from Excel, so that I don't type them by hand	A file with 200 rows imports in under 1 minute. Rows with a missing university ID or an unknown course are listed with the reason and not imported. Importing the same file again creates no duplicates	DB-03
US-04	As any user, I want to log in once and land in my own interface	A student lands in the Student interface, a supervisor in the Supervisor interface, a technician in Maintenance. A user with two roles chooses one. Every screen is in Arabic, right to left, by default	ST-01, SYS-01, SYS-07

Sprint 2 — Labs

ID	Story	Acceptance criteria	Covers
US-05	As a student, I want to add a lab case to my lab work card	The case type is chosen from the course's list, and the exercise or model number is entered. The case shows the course's steps, all pending	ST-10
US-06	As a student, I want to finish a lab step and send it with photos	At least 1 photo is required when the step template says so. Upload works on a slow network and retries after loss. The step shows as "Waiting for approval"	ST-09, ST-11
US-07	As a supervisor, I want to grade lab work from photos on my phone	The queue lists waiting steps for my courses only. An approval with a grade takes 2 taps after opening. A return needs a comment. The student is notified within 1 minute	SV-07, SYS-03
US-08	As a student, I want to see my progress on each work card	The card shows each case type as done / required, plus the grades. A case counts as done only when its last step is approved	ST-07, ST-08

Sprint 3 — Patients and diagnosis

ID	Story	Acceptance criteria	Covers
US-09	As the super admin, I want to register a patient once	The file number is generated and unique. A matching national ID blocks saving; a matching name plus phone shows a warning. The patient can be sent to a diagnosis clinic's queue	DB-08, DB-09
US-10	As a supervisor, I want to give a waiting patient to a student for diagnosis	The queue shows the patients waiting in my diagnosis clinics. After assignment, the patient appears in the student's "My patients" list	SV-03
US-11	As a student, I want to fill the diagnosis on my phone	All F1 sections are present. Tapping a tooth opens its codes, and several codes per tooth are allowed. Index scores accept only the configured range. A draft survives closing the app	ST-05
US-12	As a student, I want the patient's medical risks shown clearly	Any medical history item flagged as an alert (for example anticoagulants or allergy) shows as a red banner on every screen of that patient	ST-03, ST-04
US-13	As a supervisor, I want to approve or return a diagnosis	Approval locks it. A return needs a comment and sends it back to the student as editable. Both actions are logged with name and time	SV-04, SYS-05
US-14	As a supervisor, I want to refer plan items to departments	Only items of an approved diagnosis can be referred, and each item can be in only one open referral. The receiving clinic sees the referral at once	SV-05

Sprint 4 — Clinical cards and maintenance

ID	Story	Acceptance criteria	Covers
US-15	As a supervisor in a department clinic, I want to accept a referral and assign the patient to a student	Only eligible students are offered (enrolled in a course run in this clinic this term), with their progress shown. The case appears on the student's clinical work card with the patient, tooth and case type filled in	SV-06
US-16	As a supervisor, I want closing the last step to close the referral	Approving the last step and confirming completion marks the case done, the referral completed and the plan item done. The referring clinic is notified	SV-08
US-17	As a supervisor, I want to move a patient to another student	The open cases move to the new student's card, and completed steps stay credited to the old student. The reason and the date are recorded	SV-09
US-18	As a technician, I want to do the morning round quickly	Marking 20 chairs "All OK" takes under 1 minute. Only one round per clinic, round type and day is allowed. A fault requires a note	MT-03

ID	Story	Acceptance criteria	Covers
US-19	As a technician, I want a fault to become a ticket that I follow until it is repaired	The ticket is created from the failed check. A blocking category takes the chair out of service. Resolving it puts the chair back and records who fixed it and how	MT-04, MT-05, MT-06
US-20	As the super admin, I want to distribute students to chairs	The suggestion fills each chair with up to the course's students per chair and never uses out-of-service chairs. Changes are saved per date and session	DB-12
US-21	As a supervisor, I want to mark attendance on the chair grid	Each student in each session is marked present or absent. The daily clinic report (F8) shows it	SV-11

Sprint 5 — Reports and hardening

ID	Story	Acceptance criteria	Covers
US-22	As management, I want the daily reports without paper	F6/F18, F8 and F9 reports are available per clinic and date, as on-screen tables and as Excel and PDF exports	DB-15
US-23	As a supervisor, I want to find patients who still need a given treatment	A search by treatment type returns patients with an approved, not yet referred plan item of that type, oldest first	SV-10
US-24	As any user, I want to be told when something needs me	Every event in SYS-03 creates an in-app notification. Tapping it opens the item. The unread count shows on the home screen	SYS-03, ST-14, SV-14

8. Non-functional requirements

Each requirement has a number that can be tested. Privacy and audit are mandatory because the system holds medical records.

ID	Area	Requirement
NFR-01	Performance	API responses within 1 second for 95% of requests; patient search under 2 seconds; app start to home screen under 3 seconds on a mid-range phone
NFR-02	Capacity	300 users at the same time; 20,000 patients and 500,000 photos without redesign
NFR-03	Availability	99% during clinic hours (Sunday–Thursday, 8:00–16:30 Damascus time); maintenance outside those hours
NFR-04	Backup	Daily encrypted database backup and daily photo storage backup, kept 30 days; a restore tested before go-live
NFR-05	Security	HTTPS only; passwords hashed with Django's default (PBKDF2 or Argon2); tokens expire (access token 15 minutes, refresh token 30 days); permissions checked on the server

ID	Area	Requirement
NFR-06	Privacy	Students see only their own patients; photos are not saved to the phone's gallery after upload; the dashboard is not reachable by students
NFR-07	Audit	Every approval, change and login is logged and kept for at least 5 years
NFR-08	Usability	A normal step approval takes 2 taps; a full round of 20 chairs under 1 minute when all are OK; touch targets at least 48 px
NFR-09	Localization	Arabic (right to left) and English; dates shown day/month/year; the language can be switched at any time
NFR-10	Network	The app keeps working on a slow connection. Drafts are saved on the phone, and photo uploads queue and retry
NFR-11	Compatibility	Android 8.0 and later; iOS 15 and later; current desktop browsers for the dashboard
NFR-12	Maintainability	One repository; a code style enforced by linters; backend test coverage of at least 70% for the business rules; a CLAUDE.md that documents the conventions
NFR-13	Configurability	Every item in section 4 can be changed in the dashboard without code changes or an app release
NFR-14	Store compliance	Demo accounts for the App Store and Google Play reviewers; a privacy policy page; a permission prompt for camera access that states the reason

9. External interfaces

The system talks to four things outside itself: the mobile app (through its API), Excel files, the push notification services, and file storage. There is no live connection to the academic system or the imaging software.

Interface	Direction	Format and rules
REST API	App ↔ backend	JSON over HTTPS under `/api/v1/`; JWT in the `Authorization` header; paginated lists; errors as `{code, message, details}` with messages in the user's language; OpenAPI schema at `/api/v1/schema/`
Excel import	File → backend	.xlsx templates downloadable from the dashboard. **Students:** university ID, full name (AR), full name (EN), mobile, study year. **Enrollments:** university ID, course code, term. Validated row by row, with a report of errors
Excel and PDF export	Backend → file	Reports and record printouts (DB-15, SYS-08)
Push notifications	Backend → phone	Firebase Cloud Messaging for Android and Apple Push Notification service for iOS. If push fails, the in-app notification list still holds every event
File storage	Backend ↔ storage	S3-compatible object storage; private files served through short-lived signed links; thumbnails created on upload
X-ray images	User → backend	Uploaded as image files (JPEG or PNG) exported from the imaging software. A direct link to that software is not in scope